Deep Learning

AI Engineer Placement Preparation Series

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Why This Matters

Deep Learning has revolutionized AI. Every cutting-edge AI system—from ChatGPT to computer vision to autonomous vehicles—is built on neural networks. For AI Engineer placements in 2025-2026, you MUST understand deep learning fundamentals: how neural networks learn, what happens during backpropagation, why certain optimizers work better than others, and how to train networks that don't overfit.

2025-2026 Shift: Interviewers now expect you to not just know the theory, but to have hands-on experience. You should be able to build and debug a neural network from scratch using PyTorch. You should understand why your model is overfitting, and know 5+ techniques to fix it. You should know when to use CNNs vs RNNs vs Transformers (covered in NLP module). Knowledge of both theory AND implementation is mandatory.

1. Neural Network Fundamentals

1.1 The Perceptron

The foundation of all neural networks is the perceptron. A perceptron is a single neuron that takes multiple inputs (x■, x■, ..., x■), multiplies each by a weight (w■, w■, ..., w■), sums them up, adds a bias (b), and passes the result through an activation function:

```
output = activation(w■x■ + w■x■ + ... + w■x■ + b)
```

A single perceptron can only learn linearly separable problems. Multiple perceptrons stacked in layers form a neural network that can learn arbitrary non-linear relationships.

1.2 Network Architecture: Layers

Input Layer: Raw features (pixels, numerical values, embeddings). No computation. Just passes data forward.

Hidden Layers: Intermediate layers where computation happens. Each neuron in a hidden layer computes a weighted sum of inputs from the previous layer and applies an activation function. Multiple hidden layers allow the network to learn increasingly abstract representations. A network with 3+ hidden layers is "deep".

Output Layer: Final predictions. For classification, usually has 1 neuron per class with a softmax activation. For regression, 1 neuron with linear or no activation.

1.3 Forward Propagation

Forward propagation is the process of computing the output by passing data through the network layer by layer:

```
z¹ = W¹ · x + b¹          (linear transformation, layer 1)
a¹ = activation(z¹)      (apply activation)
z² = W² · a¹ + b²        (layer 2, using previous output)
a² = activation(z²)
...
■ = activation(z^L)      (final output)
```

This process is deterministic given weights and inputs. The network's ability to solve a task depends entirely on whether the weights are good. Learning = finding good weights.

2. Activation Functions & Loss Functions

2.1 Why Activation Functions?

If you stack only linear transformations (W·x + b), the network is still linear (composing linear functions gives a linear function). You cannot solve non-linear problems. Activation functions introduce non-linearity, enabling the network to learn complex patterns.

2.2 Common Activation Functions

Function	Formula	Range	Use Case	Pros/Cons
Sigmoid	σ(z) = 1/(1+e^-z)	[0, 1]	Binary classification (output layer)	Smooth, interpretable as probability. Prone to vanishing gradients.
Tanh	tanh(z) = (e^z - e^-z)/(e^z + e^-z)	[-1, 1]	Hidden layers (historically)	Centered at 0. Still suffers from vanishing gradients.
ReLU	max(0, z)	[0, ∞)	Hidden layers (DEFAULT)	Fast, doesn't saturate. Simple. Can have dying ReLU problem (neu
Leaky ReLU	max(0.01z, z)	[-∞, ∞)	Hidden layers	Fixes dying ReLU. Small negative slope prevents dead neurons.
GELU	z·Φ(z) (smooth)	varies	Transformers, modern models	Used in BERT, GPT. Smooth. More computationally expensive.
Softmax	e^zi / Σe^zj	[0, 1]	Multi-class classification (output)	Converts logits to probabilities. Sum = 1.

2.3 Loss Functions

Loss Function: Measures how wrong your predictions are. Lower loss = better model. The network learns by minimizing loss.

Task	Loss Function	Formula	When to Use
Binary Classification	Binary Cross-Entropy	-(y·log(p) + (1-y)·log(1-p))	2 classes. Output: sigmoid.
Multi-class	Categorical Cross-Entropy	-Σ(y_i·log(p_i))	3+ classes. Output: softmax.
Regression	MSE (Mean Squared Error)	(1/N)·Σ(y - ■)²	Continuous targets. Sensitive to outliers.
Regression	MAE (Mean Absolute Error)	(1/N)·Σ|y - ■|	Continuous targets. Robust to outliers.
Ranking/Triplet Loss	Triplet Loss	max(margin + d(anchor, neg) - d(anchor, pos), 0)	Learn embeddings. Face recognition, metric learning.

3. Backpropagation & Optimization

3.1 The Core Insight: Chain Rule

Backpropagation is the algorithm for computing gradients (derivatives of loss with respect to each weight). It uses the chain rule: if loss = f(g(h(x))), then d(loss)/dx = d(f)/dg · dg/dh · dh/dx. By computing gradients backward through the network, we efficiently get all gradients in one pass.

3.2 The Backward Pass (Simplified)

1. Forward pass: compute output and loss

2. Backward pass: compute ∂loss/∂z for output layer

3. Chain backward: ∂loss/∂z for hidden layers using chain rule

4. Weight gradients: ∂loss/∂w = ∂loss/∂z · ∂z/∂w = ∂loss/∂z · x

5. Update: w_new = w_old - learning_rate · ∂loss/∂w

3.3 Vanishing & Exploding Gradients

Vanishing Gradients: In deep networks, gradients are multiplied together layer by layer. If activation derivatives are small (<0.5 for sigmoid), gradients shrink exponentially. Early layers get near-zero gradients and barely learn. Solution: ReLU, residual connections, batch normalization.

Exploding Gradients: If weights are large or activation derivatives > 1, gradients grow exponentially. Causes NaN losses and training crashes. Solution: gradient clipping, weight initialization, normalization.

4. Optimizers Deep Dive

4.1 Gradient Descent Variants

Optimizer	Update Rule	Pros	Cons	Use Case
Batch GD	w = w - lr·∂L/∂w (all data)	Stable, exact gradient	Slow, memory heavy	Small datasets
SGD	w = w - lr·∂L/∂w (1 sample)	Fast, online learning	Noisy, diverges easily	Large datasets
SGD + Momentum	v = β·v - lr·∂L/∂w; w = w + v	Accelerates, dampens noise	Hyperparameter tuning	Medium speed
Nesterov	Look-ahead momentum	Faster convergence	Slightly more complex	CNN training
Adagrad	Adaptive per-param lr	Auto-scale learning rates	lr decreases monotonically	Sparse features
RMSProp	Adaptive lr with decay	Good for RNNs	Not as standard	RNN training
Adam	Momentum + RMSProp hybrid	Works out-of-box, best default	Can overfit on small data	DEFAULT
AdamW	Adam with decoupled weight decay	Better generalization than Adam	Slightly slower	Modern standard

4.2 Learning Rate Schedulers

Constant learning rates are rarely optimal. Start with a high learning rate to escape bad minima, then reduce it to fine-tune. Common schedulers:

Step Decay: Reduce by factor (e.g., 0.1) every N epochs.

Cosine Annealing: Smoothly decrease lr following a cosine curve. Often better than step decay.

Warmup: Increase lr from 0 to target over first N steps, then decay. Used in Transformers. Stabilizes training.

ReduceLROnPlateau: Reduce lr if validation loss stops improving. Reactive.

5. Regularization Techniques

5.1 L1 & L2 Regularization (Weight Decay)

L2 Regularization: Add $\lambda \cdot \Sigma(w^2)$ to the loss. Encourages all weights to be small. In deep learning frameworks, this is called weight_decay parameter.

L1 Regularization: Add $\lambda \cdot \Sigma(|w|)$ to the loss. Drives some weights to exactly 0. Rare in deep learning (preferred in traditional ML).

5.2 Dropout

During training, randomly set a fraction p of neurons to 0 (deactivate them). Forces the network to learn redundant representations. Acts like training an ensemble of smaller networks. Prevents co-adaptation of neurons.

```
During training: h_dropped = h * mask, where mask ~ Bernoulli(1-p)
During inference: use full network (or scale by 1-p). PyTorch handles this.
Typical p: 0.5 for fully connected, 0.1-0.3 for CNNs.
```

5.3 Batch Normalization

Normalize inputs to each layer to have mean=0, std=1 within a batch. Reduces internal covariate shift (distribution shift of layer inputs). Massive practical impact: faster training, less sensitive to initialization, acts as regularizer.

```
x_norm = (x - batch_mean) / sqrt(batch_var + ε)
y = γ·x_norm + β  (learnable scale & shift)
```

5.4 Other Regularization Techniques

Early Stopping: Monitor validation loss. Stop training if it stops improving for N epochs. Prevents overfitting.

Data Augmentation: Rotate, crop, flip, brightness adjust images. Increases dataset size without more data collection.

Layer Normalization: Like batch norm but normalizes per sample (not per batch). Better for RNNs and Transformers. No batch dependency.

6. Convolutional Neural Networks (CNNs)

6.1 Why CNNs for Images?

Fully connected layers treat all pixels equally. For a 224×224 RGB image, that's 150K input features → completely unscalable and ignores spatial structure. CNNs exploit two properties of images: (1) local patterns matter (a cat's ear is a local pattern), (2) patterns repeat (what you learn at top-left should apply to bottom-right). This reduces parameters dramatically and works much better.

6.2 Convolution Operation

A conv filter (kernel) is a small learned weight matrix (e.g., 3×3). Slide it across the image, computing element-wise products and summing. This creates a new feature map. Multiple filters → multiple feature maps.

```
Filter example (3x3):
[-0.1,  0.2,  0.5]
[ 0.3, -0.2,  0.1]
[ 0.0,  0.4, -0.3]

Slide across image. At each position:
output = sum(filter * input_patch)

16 filters → 16 feature maps, each learning different patterns
```

6.3 Key CNN Components

Stride: Step size when sliding the filter. stride=1 (standard) slides 1 pixel. stride=2 jumps 2 pixels. Larger stride = smaller output. Speeds up computation.

Padding: Add zeros around image edges. 'same' padding keeps dimensions the same. 'valid' (no padding) shrinks output. Without padding, edges are undersampled.

Pooling: Downsampling. Max pooling: take max of 2×2 window (or other size). Average pooling: take mean. Reduces spatial dimensions, increases receptive field, adds robustness.

6.4 Classic CNN Architectures

Architecture	Year	Key Innovation	Accuracy (ImageNet)	When to Use
LeNet	1998	First CNN. 7 layers.	N/A	Educational only
AlexNet	2012	Deep CNN. ReLU. Dropout.	~63%	Historical
VGGNet	2014	Small 3×3 filters. Very deep (16-19 layers).	~71%	Baseline
ResNet	2015	Skip connections. 50-152 layers. SOTA.	~77%	Industry standard
DenseNet	2016	Dense skip connections. Efficient.	~77%	When memory is tight
EfficientNet	2019	Efficient scaling. Adjustable size/speed/accuracy.	~84%	Production: tunable

6.5 Transfer Learning

Pre-trained models (trained on ImageNet with millions of images) have learned general visual features (edges, textures, shapes, parts). You can reuse these and fine-tune on your task. Strategy:

1. Load pre-trained weights (e.g., ResNet50 from PyTorch Hub)

2. Replace final classification layer with your problem's output size

3. Option A (fast): Freeze early layers, train only final layers

4. Option B (better): Unfreeze all, train with very low learning rate

Result: Need 10-100x less data, much faster training, better performance. This is the standard in production.

7. Recurrent Neural Networks (RNNs)

7.1 Why RNNs for Sequences?

Text, audio, time-series are sequential: each word/sample depends on previous ones. RNNs maintain a hidden state that persists across time steps, allowing them to remember context. Unlike CNNs (spatial), RNNs process temporal/sequential dependencies.

7.2 Vanilla RNN

```
At each time step t:
h_t = activation(W_h·h_{t-1} + W_x·x_t + b)
y_t = W_y·h_t + c

h_t: hidden state (carries memory)
x_t: input at time t
W_h: recurrent weight matrix (applied to previous hidden state)
W_x: input weight matrix
```

Problem: Long-term dependencies are lost. Gradients vanish over long sequences (same issue as deep networks). The network 'forgets' early inputs.

7.3 LSTM (Long Short-Term Memory)

LSTMs solve vanishing gradients with a cell state (separate from hidden state) and gates that control information flow.

```
Forget gate: f_t = sigmoid(W_f·[h_{t-1}, x_t] + b_f)  (what to forget)
Input gate: i_t = sigmoid(W_i·[h_{t-1}, x_t] + b_i)   (what to remember)
Cell candidate: C■_t = tanh(W_c·[h_{t-1}, x_t] + b_c)  (new information)
Cell state: C_t = f_t ■ C_{t-1} + i_t ■ C■_t          (maintain long-term memory)
Output gate: o_t = sigmoid(W_o·[h_{t-1}, x_t] + b_o)  (what to output)
Hidden state: h_t = o_t ■ tanh(C_t)
```

Key insight: Cell state has additive updates (+ not ×), so gradients don't vanish. Information can flow undisturbed over long sequences.

7.4 GRU (Gated Recurrent Unit)

Simpler than LSTM. Combines forget and input gates into a single "reset gate." Similar performance but fewer parameters. Often preferred when training data is limited.

7.5 Bidirectional RNNs

Process sequence left-to-right AND right-to-left. Concatenate both hidden states. Useful when you have full sequence (not generating next token). Example: NER, classification. Not for auto-regressive generation (you can't look at future tokens).

8. PyTorch (Industry Standard)

8.1 Core Concepts

```
import torch
import torch.nn as nn
import torch.optim as optim

# Tensors (PyTorch's arrays)
x = torch.randn(32, 28, 28)  # batch of 32 28x28 images
x.shape                       # torch.Size([32, 28, 28])
x.to('cuda')                  # move to GPU

# Autograd: automatic differentiation
x = torch.tensor([2.0], requires_grad=True)
y = x ** 2 + 3*x
y.backward()     # compute gradients
print(x.grad)    # dy/dx = 4x+3 = 11 at x=2
```

8.2 Building a Network

```
class SimpleNet(nn.Module):
    def __init__(self):
        super().__init__()
        self.fc1 = nn.Linear(784, 128)      # 28×28 → 128
        self.fc2 = nn.Linear(128, 64)       # 128 → 64
        self.fc3 = nn.Linear(64, 10)        # 64 → 10 classes
        self.dropout = nn.Dropout(0.2)

    def forward(self, x):
        x = x.view(x.size(0), -1)           # flatten to [batch, 784]
        x = torch.relu(self.fc1(x))
        x = self.dropout(x)
        x = torch.relu(self.fc2(x))
        x = self.fc3(x)
        return x  # logits, NOT softmax yet!

model = SimpleNet()
criterion = nn.CrossEntropyLoss()  # combines softmax + cross-entropy
optimizer = optim.Adam(model.parameters(), lr=0.001)
```

8.3 Training Loop

```
for epoch in range(10):
    for X_batch, y_batch in train_loader:
        # Forward pass
        logits = model(X_batch)
        loss = criterion(logits, y_batch)

        # Backward pass
        optimizer.zero_grad()  # clear old gradients
        loss.backward()         # compute gradients
        optimizer.step()        # update weights
```

```
    # Validation
    with torch.no_grad():  # disable gradient computation
        val_loss = 0
        for X_val, y_val in val_loader:
            val_loss += criterion(model(X_val), y_val)
        print(f'Epoch {epoch}, Val Loss: {val_loss/len(val_loader):.4f}')
```

8.4 CNN in PyTorch

```
class CNN(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv1 = nn.Conv2d(3, 32, kernel_size=3, padding=1)
        self.pool = nn.MaxPool2d(2)
        self.conv2 = nn.Conv2d(32, 64, kernel_size=3, padding=1)
        self.fc = nn.Linear(64*56*56, 10)  # adjust size based on input

    def forward(self, x):
        x = self.pool(torch.relu(self.conv1(x)))  # [B,3,224,224]→[B,32,112,112]
        x = self.pool(torch.relu(self.conv2(x)))  # [B,32,112,112]→[B,64,56,56]
        x = x.view(x.size(0), -1)               # flatten
        x = self.fc(x)
        return x
```

8.5 Transfer Learning in PyTorch

```
import torchvision.models as models

# Load pre-trained ResNet50
model = models.resnet50(pretrained=True)

# Freeze early layers (optional, for fast training)
for param in model.parameters():
    param.requires_grad = False

# Replace final layer for your task (10 classes)
model.fc = nn.Linear(2048, 10)

# Only train the new layer
optimizer = optim.Adam([p for p in model.parameters() if p.requires_grad], lr=0.001)
# OR unfreeze all and train with low lr
for param in model.parameters():
    param.requires_grad = True
optimizer = optim.Adam(model.parameters(), lr=0.00001)
```

9. TensorFlow / Keras

9.1 Why Keras?

Keras is a high-level API (now integrated into TensorFlow 2.x). Simpler syntax than PyTorch for simple models, but less flexible for custom training loops. Many tutorials and papers use Keras. Know both PyTorch and Keras.

9.2 Keras Workflow

```
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers

# Build sequential model
model = keras.Sequential([
    layers.Dense(128, activation='relu', input_shape=(784,)),
    layers.Dropout(0.2),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])

# Compile
model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

# Train
model.fit(X_train, y_train_onehot, epochs=10, validation_split=0.2, batch_size=32)

# Evaluate
model.evaluate(X_test, y_test_onehot)

# Predict
predictions = model.predict(X_new)
```

9.3 Functional API (More Powerful)

For complex architectures (multi-input, multi-output, skip connections), use Functional API instead of Sequential:

```
inputs = keras.Input(shape=(28, 28, 1))
x = layers.Flatten()(inputs)
x = layers.Dense(128, activation='relu')(x)
x = layers.Dense(64, activation='relu')(x)
outputs = layers.Dense(10, activation='softmax')(x)

model = keras.Model(inputs=inputs, outputs=outputs)
```

9.4 Key Difference: PyTorch vs TensorFlow

Aspect	PyTorch	TensorFlow/Keras
Learning Curve	Easier, Pythonic	Steeper initially
Debugging	Easier (imperative)	Harder (graph-based)
Research	Preferred by researchers	Production at scale
Community	Very strong in academia	Mature, large industry use
Flexibility	More flexible, custom training	Simpler for standard tasks

10. Common Mistakes

Not using batch normalization in deeper networks: BN stabilizes training dramatically. Add it after conv/dense layers.

Training on GPU but not moving model to GPU: model.to('cuda') is easy to forget. Check device mismatch errors.

Applying softmax in loss function AND in model output: Use CrossEntropyLoss (includes softmax). Don't double-apply.

Forgetting optimizer.zero_grad(): Gradients accumulate by default. Always zero before backward().

Using model.eval() or torch.no_grad() during training: eval() disables dropout/batch norm. Only for validation/test.

High learning rate causing NaN loss: Start with lr=0.001 or 0.0001. If loss NaNs, reduce. Use learning rate schedulers.

Not shuffling training data: Shuffling is critical for convergence. DataLoaders do this by default (shuffle=True).

Using entire dataset as validation: Wastes training data. Use 10-15%. Or use k-fold cross-validation.

Not normalizing input data: Normalize images to [0,1] or standardize to mean=0, std=1. Dramatically speeds up training.

Training too deep networks without skip connections: Deep networks (50+ layers) need residual/skip connections to avoid vanishing gradients.

11. Quick-Fire Q&A;

[Easy] What is backpropagation?

Algorithm to compute gradients of loss w.r.t. weights using chain rule. Efficient: computes all gradients in one backward pass.

[Easy] Why use ReLU over sigmoid?

ReLU: fast, doesn't saturate. Sigmoid: gradients near 0/1 are tiny (vanishing gradients). ReLU is standard in hidden layers.

[Medium] Explain batch normalization.

Normalize layer inputs to mean=0, std=1. Reduces internal covariate shift, allows higher learning rates, speeds training. Use with dropout carefully.

[Medium] What's the difference between dropout and batch norm?

Dropout: zeros random neurons during training (regularizer). Batch norm: normalizes activations (accelerator + regularizer). Both help generalization.

[Medium] When should you use LSTM vs GRU vs vanilla RNN?

Vanilla RNN: simple sequences only. GRU: good balance of power/simplicity. LSTM: long sequences, more capacity. Transformer: now preferred for most NLP.

[Hard] How does the kernel trick work in CNNs?

Convolution is a learned filter. Parameter sharing (same filter applied across space) drastically reduces parameters vs fully connected.

[Medium] What is transfer learning and when to use it?

Reuse pre-trained weights, fine-tune on new task. Use when: limited labeled data, similar domain, compute budget matters. Almost always helps.

[Medium] Explain vanishing gradients.

In deep networks, gradients $\propto$ product of derivatives. If derivatives <1, product $\rightarrow$ 0. Early layers get ~zero gradient, can't learn. Fix: ReLU, batch norm, skip connections.

[Easy] What's the difference between training and inference?

Training: forward pass, backward pass, gradient updates. Inference: forward only, no gradients. Use model.eval() in PyTorch, disable dropout/batch norm updates.

[Medium] How do you prevent overfitting in deep networks?

More data, dropout, regularization (L2), early stopping, data augmentation, simplify model, batch normalization. Use validation set to detect overfitting.

12. One-Line Cheat Sheet

• Backpropagation — Compute gradients via chain rule, backward through network, update weights using gradients.

• ReLU — max(0, x). Default activation. Fast, avoids vanishing gradients.

• Softmax — Converts logits to probabilities. Use for multi-class output layer.

• Sigmoid — Squeeze to [0,1]. Use for binary classification output or gates (LSTM).

• Cross-Entropy Loss — Standard for classification. Penalizes wrong confident predictions.

• MSE Loss — Standard for regression. Penalizes large errors more.

• Batch Normalization — Normalize inputs to [mean=0, std=1]. Accelerates training, regularizes.

• Dropout — Randomly deactivate neurons during training. Prevents overfitting. No effect at inference.

• Adam Optimizer — Momentum + adaptive lr. Default optimizer. Works well out-of-box.

• Learning Rate Scheduler — Reduce lr over time. Cosine annealing, step decay, warmup.

• Convolution — Slide learned filter across image. Parameter sharing. Local connections.

• Pooling — Max or average pooling. Downsamples. Adds robustness.

• Transfer Learning — Load pre-trained weights, fine-tune. Use almost always.

• LSTM — RNN variant with cell state + gates. Handles long sequences. Better than vanilla RNN.

• Bidirectional RNN — Process sequence both directions. For tasks with full context.

• PyTorch Tensors — Multi-dimensional arrays. support autograd. Move to GPU with .to('cuda').

• PyTorch nn.Module — Base class for models. Define __init__ and forward().

• DataLoader — Batches data, shuffles, allows parallel workers. Essential for efficiency.

• model.eval() — Disable dropout, batch norm updates. Use during validation/inference.

• torch.no_grad() — Disable gradient computation. Use during validation to save memory.

Final Note

Deep Learning is both theory and practice. Understand the math (gradients, activations, loss functions) but spend most time coding. Build projects: image classification, sentiment analysis, time-series forecasting. Read papers (ArXiv). Stay current—the field moves fast. For 2025-2026 placements, interviewers expect you to code a neural network from scratch AND explain why it works.